

DHRUV JYOTI DAS

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EDUCATION

SRM UNIVERSITY OF SCIENCE AND TECHNOLOGY | B. TECH CSE | CGPA: 9.0 | EXPECTED: 2027
IIT MADRAS | BS IN DATA SCIENCE | EXPECTED: 2027

TECHNICAL SKILLS

Programming: Java, C, Python, JavaScript

Web Development: Node.js, Express.js, React Native, HTML5, CSS3, WebSocket, REST API

Databases & Tools: MySQL, PostgreSQL, MongoDB, Firebase, Render, Serveo, ngrok, Git/GitHub, Tableau

Data Science & ML: Pandas, NumPy, Scikit-learn, Vision Transformers, Matplotlib, Seaborn, Gabor Filtering

AI & LLMs: RAG pipelines, multi-turn CQG, DeepSeek, local LLMs, prompt structuring, WebRTC

EXPERIENCE

SAMSUNG R&D RESEARCH INTERN

Mar 2025 – Jan 2026

- Built memory-augmented conversational system to resolve vague user queries and reduce clarification turns.
- Implemented NER ensemble with Gemini validation for domain knowledge graph construction.
- Developed minimal clarifying question generator to improve query completeness.
- Produced analytics and lightweight deployments to improve reliability and reduce serving cost.

IIT INDORE — RESEARCH INTERN (STCLN FOR CROP MAPPING)

Nov 2025 – Present

- Extended STCLN for masked self-supervised spatiotemporal learning on Sentinel-2 time-series to improve low-label crop mapping; redesigned encoder/decoder with multi-scale tokenizers, temporal cross-attention, and spectral gating.
- Ran controlled experiments with masking ratios and 1–10% labelled regimes, logging via Weights & Biases and analysing representations using CCA/t-SNE.
- Achieved +8–12% F1-macro gains in low-label settings with reduced temporal reconstruction drift and lower MSE/MAE.

PROJECTS

[\[Link\]](#) CHAT X – REAL-TIME CHAT & AI VIDEO APP

- Real-time chat backend with auth sessions, channel routing, MongoDB persistence, and sub-200 ms latency.
- Full WebRTC pipeline (ICE/SDP/re negotiation) with self-hosted STUN/TURN on Oracle VPS for reliable P2P NAT traversal.
- Transformer-based chat summarization with sentiment tagging and cross-device sync via WebSocket signalling.

[\[Link\]](#) TLS-MIMICKING TROJAN PROXY FOR CENSORSHIP RESISTANCE

- Deployed Trojan over TLS/443 on Oracle Free Tier VPS with Let's Encrypt certificates, creating an HTTPS-mimicking proxy that bypassed campus firewall DPI and SNI filtering.
- Configured v2rayN (SOCKS5), Oracle ingress rules, and Linux routing/NAT to enable stable outbound TCP/HTTPS tunnelling across restricted networks.
- Verified seamless access to blocked web services and SSH-over-TLS; documented protocol limitations (UDP-blocked gaming, QUIC filtering) under real campus network constraints.

[\[Link\]](#) CET-ViT: CAUSAL EMERGENCE TRANSFORMER FOR VISION

- Designed V-CEO, a sparse EI-guided token assignment module enabling per-image dynamic macro-region discovery without segmentation supervision; achieved 76.54% accuracy and 76.25% F1-macro on CIFAR-100, surpassing no-EI baseline by +2.39% OA and +2.58% F1.
- Formulated a two-term EI loss (entropy floor + reversibility ceiling) stabilizing CEP reversibility within [0.20, 0.45] for all 300 epochs — resolving rank-1 collapse and over-diffusion failure modes of prior designs.
- Trained from scratch on AMD MI300X using PyTorch 2.9.0 + ROCm; dynamic K estimation via Gumbel-SoftMax at 0.1% parameter overhead with 11,433 img/s throughput.

CERTIFICATIONS

ORACLE CERTIFIED PROFESSIONAL JAVA SE 11 DEVELOPER

OCI 2025 CERTIFIED DATA SCIENCE PROFESSIONAL

NPTEL ML, JAVA, NLP & DBMS